

Java Questions

Predefined Functions & User Defined Functions in Java

1. Predefined Function – String Length Validation

Question

You are developing a login system.

Check whether the entered username contains at least 8 characters using a predefined Java function.

Sample Output

Enter Username: Selva123

Valid Username

2. Predefined Function – Convert to Uppercase

Question

A company stores employee IDs in uppercase only.

Convert the given employee name into uppercase using a predefined function.

Sample Output

Enter Employee Name: selva kumar

Output: SELVA KUMAR

3. Predefined Function – Remove Extra Spaces

Question

A banking application receives customer names with unwanted spaces.

Remove starting and ending spaces using a predefined function.

Sample Output

Input: " Rahul Sharma "

Output: "Rahul Sharma"

4. Predefined Function – Find Maximum Salary

Question

An HR application needs to identify the highest salary between two employees using a predefined Java function.

Sample Output

Employee 1 Salary: 45000

Employee 2 Salary: 52000

Highest Salary: 52000

5. Predefined Function – Random OTP Generation

Question

Generate a random OTP for mobile verification using predefined Java functions.

Sample Output

Generated OTP: 483921

User Defined Function Questions

6. User Defined Function – Student Grade Calculator

Question

Create a user defined function called calculateGrade() to display student grades based on marks.

Sample Output

Enter Marks: 87

Grade: A

7. User Defined Function – Simple Interest Calculator

Question

Create a function named calculateInterest() to calculate simple interest.

Sample Output

Principal Amount: 10000

Rate of Interest: 5

Time: 2

Simple Interest: 1000.0

8. User Defined Function – Even or Odd Checker

Question

Create a user defined function called `checkEvenOdd()` to identify whether a number is even or odd.

Sample Output

Enter Number: 14

14 is Even

9. User Defined Function – Employee Bonus Calculation

Question

A company gives a 10% bonus to employees.

Create a function `calculateBonus()` to calculate the bonus amount.

Sample Output

Employee Salary: 50000

Bonus Amount: 5000

10. User Defined Function – Password Validation

Question

Create a function `validatePassword()` to check whether the password length is greater than 8 characters.

Sample Output

Enter Password: Java@1234

Password is Valid

11. User Defined Function – Electricity Bill Calculator

Question

Create a function `calculateBill()` to calculate electricity bill amount based on units consumed.

Sample Output

Enter Units: 150

Electricity Bill: 750

12. User Defined Function – Shopping Discount System

Question

An e-commerce application provides a 20% discount for purchases above ₹5000.
Create a function `calculateDiscount()`.

Sample Output

Purchase Amount: 7000

Discount Amount: 1400

Final Amount: 5600

13. User Defined Function – Attendance Eligibility

Question

Create a function `checkEligibility()` to verify whether a student is eligible for exams based on attendance percentage.

Sample Output

Attendance Percentage: 82

Eligible for Exam

14. User Defined Function – Temperature Converter

Question

Create a function `convertToCelsius()` to convert Fahrenheit temperature into Celsius.

Sample Output

Enter Fahrenheit: 98.6

Celsius: 37.0

15. User Defined Function – ATM Balance Check

Question

Create a function `withdrawAmount()` to withdraw money from an ATM account after checking balance availability.

Sample Output

Available Balance: 10000

Withdraw Amount: 2500

Transaction Successful

Remaining Balance: 7500